

Test 6: Spacing chi^2 histogram test

Verdict: PASS | Generated: 2026-09-14 23:53:15 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

?? binned comparison of spacing histogram to Wigner-Dyson. Binning choices are part of the tolerance system.

METHOD: ?? binned comparison of spacing histogram to Wigner-Dyson. Binning choices are part of the tolerance system.

Result

Test 6 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS

`plots/plot\_01.png` ? 1600×1000, ABPlotV23 engine (supersampled)

Plots

- `plots/animation.gif` ? animated reveal of every plot of this test

Reproduce with: `julia ab_cloud_v23.jl --test 6 --no-two-pass`

Independent verification guide

Every computation is logged in `logs/computation_log.txt` (timestamps + deltas); raw console output in `logs/stdout_capture.txt`.

(no log_comp entries)

FULL COMPUTATION LOG

??

CONSOLE CAPTURE

TEST 6 (secondary, HARDCORE): Objection 2: GUE Spacing Statistics ? bin-stability audit

??

C1 bins= 300: ??/df = 2.3771, p = 0.000000 ? REJECT

C1 bins= 150: ??/df = 3.6171, p = 0.000000 ? REJECT

C1 bins= 75: ??/df = 6.3149, p = 0.000000 ? REJECT

C1 bins= 37: ??/df = 11.3303, p = 0.000000 ? REJECT

C2 halves : ??/df first=2.5187, second=2.1023, rel.dev=16.5% (limit 50%)

??

C1 bin sweep PASS ? 0 flips vs majority (REJECT)

C2 halves PASS ? rel.dev = 16.5%

Test 6 [HARDCORE pass 2]: 2 sub-checks, 0 failed ? PASS